

The maximum Laplacian ratio of a tree for all $n \geq 303$: concave witnesses and one-variable certificates

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Abstract

For a tree T with Laplacian $L(T)$ the *Laplacian ratio* is $\pi(T) = \text{per } L(T) / \prod_v \deg v$. Brualdi and Goldwasser asked in 1984 which tree on n vertices maximises it. A recent preprint of Murphy answers the question for every n , with a proof checked by the Lean kernel whose computational parts are large families of multi-dimensional certificates. We give a self-contained proof for every $n \geq 303$ in which each certificate is an inequality in one real variable. The tool is a potential, in the sense of ergodic optimisation, that depends only on the cavity message of a branch and is *concave* in it. Jensen’s inequality then replaces the children of a vertex by their mean message, and every per-vertex inequality in the argument, namely the sharp ceiling on pendant subtrees, the linear-deficit cells under a degree cap, the root bounds and the high-degree envelope, becomes a statement in two variables that is checked by tangent lines on cells, in exact rational arithmetic with the transcendental constants kept symbolic. The reduction to spiders holds for $n \geq 303$, against 492 in the preprint. The optimisation over spiders is done for $n \geq 492$ by a tangent bound, three exchanges and six polynomial inequalities, and for $303 \leq n \leq 491$ by an exact enumeration of balanced spiders.

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1 Introduction

1.1 The problem

Let T be a tree on $n \geq 2$ vertices with adjacency matrix A and diagonal degree matrix D , and let $L(T) = D - A$ be its Laplacian. Since the permanent $\text{per } M = \sum_{\sigma} \prod_i M_{i\sigma(i)}$ of the Laplacian grows with the degrees, Brualdi and Goldwasser [1] normalised it and studied

$$\pi(T) = \frac{\text{per } L(T)}{\prod_{v \in V(T)} \deg v}.$$

They proved $\pi(T) \geq 2$ with equality exactly for the star, and asked which tree on n vertices maximises π . The permanent itself is largest for the path; the normalisation removes that trivial effect and produces a problem with a rigid and surprising answer.

The answer was found by Murphy [2] for every $n \geq 4$, after a conjecture of Wu, Dong and Lai [5] for subdivided stars had been refuted by Pant [3]. It is built from two small pieces. A *cherry* is a pendant path of length two. For $j \geq 1$ an *arm* A_j is a vertex carrying j cherries and nothing else, attached to a further vertex; it has $2j + 1$ vertices, and A_0 denotes a leaf. A *spider* is a tree with a vertex, its *centre*, all of whose branches are leaves, cherries or arms; we write $C^c A_{j_1} A_{j_2} \cdots$ for the spider whose centre carries c cherries and the arms listed, with exponents for repetitions. For large n the maximiser is a spider whose centre carries only arms, almost all of them A_5 , the exceptions being dictated by n modulo 11.

1.2 Results

Theorem 1.1 (main theorem). *Let $n \geq 303$.*

(a) *If $n \geq 492$, put $s = 6(n - 1) \bmod 11$. The unique tree on n vertices maximising π is the spider whose centre carries only arms, all of them A_5 except that*

- *for $s \in \{1, 2\}$, and for $s = 3$ with $n \geq 733$, and for $s = 4$ with $n \geq 2330$, exactly s arms are A_6 ;*
- *for $s \in \{5, \dots, 10\}$, and for $s = 3$ with $n \leq 722$, and for $s = 4$ with $n \leq 2319$, exactly $11 - s$ arms are A_4 .*

(b) *If $303 \leq n \leq 491$, the unique maximiser is the spider listed in Appendix C.*

The recipe of (a) is easy to remember: an A_5 has 11 vertices, an A_4 has 9 and an A_6 has 13; after the centre there are $n - 1 \equiv 2s \pmod{11}$ vertices to place, and s copies of A_6 , or $11 - s$ copies of A_4 , absorb the remainder. The theorem says which of the two ways wins, and the two late switches are where two very close competitors trade places. The statement is exactly the one proved in [2]; what is new here is the proof, which is organised around the following two results.

Theorem 1.2 (reduction to spiders). *For every $n \geq 303$, every tree on n vertices that maximises π is, after rerooting at a vertex of maximum degree, a spider.*

Theorem 1.3 (sharp ceiling). *Let $\rho = (621/64)^{1/11} \approx 1.229474$. Every pendant subtree b of a tree, with the degree of its root counted in the whole tree, has weighted matching sum $T_b \leq \rho^{|b|}$, with equality if and only if b is the five-cherry arm A_5 . Consequently $\max_{|T|=n} \pi(T) \leq 2\rho^{n-1}$.*

Theorem 1.3 is Theorem 4 of [2]; Theorem 1.2 is Theorem 9 there for $n \geq 492$. Our contribution is a different and much smaller verification of both, and of the optimisation over spiders behind Theorem 1.1(a).

1.3 The ideas

Local profits. For a tree, the permanent of the Laplacian is a weighted count of matchings (Theorem 2.1), and a cavity recursion computes it from the leaves up (Proposition 2.3). Each pendant subtree b summarises itself by a weight T_b and a *message* $y_b \in (0, 1]$, and $\log T_b$ is a sum over the vertices of b of a local term that depends only on the degree of the vertex and the sum of the messages of its children. Writing $F^* = \log \rho$ and $\ell(b) = \log T_b - |b|F^*$ for the *profit* of a branch, the whole problem is to understand profits.

Potentials. A standard way to bound a sum of local terms along a tree is a potential, in ergodic optimisation a *subaction*: a function $\varrho \geq 0$ of the local state such that every vertex can pay its local profit out of the credit banked by its children and still bank credit for its parent. Summing over the tree, all credits cancel along edges and the profit is at most minus the credit of the root. This is how [2] proves Theorem 1.3, with a potential depending on the number of children of a vertex and on its message, verified by cells indexed by the types of all children.

Concave potentials and Jensen. The observation at the heart of this paper is that the potential can be taken to be a *concave function of the message alone*. Then, at a vertex with m children, the total credit of the children is at most m times the potential at their mean message, and the inequality to be verified involves only two numbers, the number of children and the mean message. All the other per-vertex inequalities of the argument are of the same kind: the linear-deficit cells that drive the reduction to spiders for small centre degree, the root bounds that turn them into a threshold, and the envelope for large centre degree. For these, atoms must be exempt, so leaf and cherry children are treated exactly and only arms and non-atom branches are pooled; and for the envelope the bound carried by a branch and the function used for pooling must be allowed to differ (Lemma 3.2). Every certificate is then a tangent line on a cell of the real line, and the whole verification of the reduction takes minutes.

Exact certificates. All numbers in the certificates are rational combinations of logarithms of rationals. They are kept symbolic, so that the forced equalities at the leaf, the cherry and A_5 cancel exactly, and every other value is bounded by a rational using enclosures of $\log q$ from the series of $2 \operatorname{artanh} \frac{q-1}{q+1}$ with an explicit tail. No floating point is trusted anywhere.

Spiders by exchanges. Over spiders the value has a closed form. A tangent bound prices each departure from the all- A_5 spider, three local exchanges remove cherries and excess arms, and what is left is a comparison of two shapes in each residue class of n modulo 11, a polynomial inequality in one integer variable.

A probabilistic representation. Let every vertex of a tree pick a uniformly random neighbour and let X be the number of edges chosen from both ends. Then $\pi(T) = \mathbb{E} 2^X$ (Remark 2.2). This is not used in the proof, but it makes the lower bound of Brualdi and Goldwasser a one-line consequence of Jensen’s inequality and explains the cavity quantities.

1.4 Relation to the preprint and organisation

We reuse from [2] the matching-sum formula, the cavity recursion, the closed form and balance lemma for spiders, the benchmark spider and the answer itself. Everything else is proved afresh, with different certificates. The preprint covers all $n \geq 4$; for $150 \leq n \leq 491$ it uses credited cells with per-size knapsacks and for $7 \leq n \leq 149$ Bellman envelope tables, and we do not reproduce either, which is why our result starts at 303. Appendix A compares the two arguments component by component.

Section 2 collects the preliminaries. Section 3 develops concave witnesses and the certificate format. Sections 4 to 7 prove the four per-vertex statements, and Section 8 the reduction theorem. Section 9 optimises over spiders and Section 10 assembles the main theorem. Section 11 describes the computations and Section 12 what a formalisation would have to check.

2 Preliminaries

2.1 Permanents and matchings

Theorem 2.1 ([1, 2]). *For a tree T ,*

$$\pi(T) = \sum_M \prod_{uv \in M} \frac{1}{\deg u \deg v},$$

the sum over all matchings M of T , the empty one included.

Proof. A term $\prod_i L_{i\sigma(i)}$ of the permanent is nonzero only if every i is fixed by σ or sent to a neighbour, so the non-trivial cycles of σ run along edges of T ; a tree has no cycles of length three or more, so they are transpositions along edges, that is, a matching. A fixed point contributes $\deg i$ and a matched edge contributes $(-1)(-1) = 1$. Divide by $\prod_v \deg v$. $\square$

Remark 2.2 (a probabilistic representation). Let every vertex v choose a neighbour $c(v)$ uniformly at random, independently, and let X be the number of edges uv with $c(u) = v$ and $c(v) = u$. The mutually chosen edges form a matching $\mathcal{M}$, and $2^X = \sum_{M \subseteq \mathcal{M}} 1$, so $\mathbb{E} 2^X = \sum_M \Pr(M \subseteq \mathcal{M})$. For a matching M the events $\{c(u) = v, c(v) = u\}$, $uv \in M$, involve disjoint sets of vertices and are independent, each of probability $1/(\deg u \deg v)$. Hence $\pi(T) = \mathbb{E} 2^X$. Jensen's inequality gives $\pi(T) \geq 2^{\mathbb{E} X} = 2^{R(T)}$ with $R(T) = \sum_{uv \in E} 1/(\deg u \deg v)$ the general Randić index with exponent -1 , and adding leaves one at a time shows $R(T) \geq 1$ with equality only for the star. This is the lower bound of Brualdi and Goldwasser, slightly strengthened.

2.2 The cavity recursion

A *planted branch* b is a rooted tree whose root will be attached to one further vertex, its parent; the degree of the root is $d_b = (\text{number of children}) + 1$, and all other degrees are as in b . Let T_b be the matching sum of Theorem 2.1 restricted to the edges of b and T_b° the same sum over matchings that leave the root unmatched, and put

$$y_b = \frac{T_b^\circ}{d_b T_b} \in (0, 1],$$

the *message* of b . A leaf has $T = 1$ and $y = 1$.

Proposition 2.3 (cavity recursion). *If the root of b has children $c_1, \dots, c_m$, then with $d = m + 1$ and $R = \sum_i y_{c_i}$,*

$$T_b = \left(\prod_i T_{c_i} \right) \frac{d + R}{d}, \quad y_b = \frac{1}{d + R}. \quad (1)$$

If a tree is rooted at a vertex with k children $b_1, \dots, b_k$ and $S = \sum_i y_{b_i}$, then

$$\pi(T) = \left(\prod_i T_{b_i} \right) \left(1 + \frac{S}{k} \right). \quad (2)$$

Proof. Split the matchings of b by whether the root is unmatched, contributing $\prod_i T_{c_i}$, or matched to the child c_i , contributing $\frac{1}{d} T_{c_i}^\circ \prod_{j \neq i} T_{c_j} = \frac{y_{c_i}}{d} \prod_j T_{c_j}$. Sum. For y_b note $T_b^\circ = \prod_i T_{c_i}$. At the root of a tree the same computation applies with $d = k$. $\square$

Taking logarithms, $\log T_b$ is a sum over the vertices v of b of $\log(1 + R_v/d_v)$.

2.3 Atoms, profits and spiders

The *atoms* are the leaf, the cherry and the arms A_j , $j \geq 1$. From (1):

atom	vertices	weight T	message y
leaf	1	1	1
cherry	2	$3/2$	$1/3$
arm A_j	$2j + 1$	$(3/2)^j \frac{4j + 3}{3(j + 1)}$	$\frac{3}{4j + 3}$

In particular $T(A_5) = 621/64$. We write $\alpha_j = \frac{4j+3}{3(j+1)}$, $b_j = \frac{3}{4j+3}$, $y_{\text{ch}} = \frac{1}{3}$ and $y^* = b_5 = \frac{3}{23}$, and

$$F^* = \frac{1}{11} \log \frac{621}{64} \approx 0.2065862, \quad \rho = e^{F^*}, \quad \ell(b) = \log T_b - |b|F^*,$$

so that $\ell(A_5) = 0$. The *defects* $\lambda_j = -\ell(A_j)$ and $\varepsilon_0 = -\ell(\text{cherry}) = 2F^* - \log \frac{3}{2} \approx 0.007707$ will be used throughout; $\ell(\text{leaf}) = -F^*$.

Lemma 2.4. $\lambda_5 = 0$ and $\lambda_j > 0$ for every $j \neq 5$; that is, A_5 is the unique arm of largest weight per vertex. Numerically:

j	1	2	3	4	5	6	7	8	9	10	11	12	13
λ_j	.0601	.0213	.00656	.00103	0	.00152	.00460	.00873	.0136	.0190	.0247	.0308	.0371

Proof. For $j \leq 70$ this is a finite check with rigorous enclosures. For $j \geq 70$, $\log T(A_j) \leq j \log \frac{3}{2} + \log \frac{4}{3}$, so the weight per vertex is at most $(j \log \frac{3}{2} + \log \frac{4}{3})/(2j+1)$, which decreases in j and at $j = 70$ is below F^* . $\square$

A spider is described, up to isomorphism, by the multiset of its centre's children. If the centre has D children with weights T_a and messages y_a , then by (2)

$$\log \pi - (n-1)F^* = \sum_a \ell(a) + \log \left(1 + \frac{S}{D}\right), \quad S = \sum_a y_a. \quad (3)$$

The *bonus* $\log(1 + S/D)$ depends only on the mean message of the children.

Lemma 2.5 (benchmark, [2]). *Let $n \geq 91$, $q = 5(n-1) \bmod 11$ and $p = (n-1-9q)/11$. The spider $A_4^q A_5^p$ has*

$$\log \pi - (n-1)F^* \geq B := 10 \ell(A_4) + \log \frac{26}{23} > 0.11233.$$

Proof. $p \geq 0$ because $9q \leq 90 \leq n-1$. By (3) the left side is $q\ell(A_4) + \log(1 + S/D)$ with $S/D \geq y^*$, since $b_4 > b_5$, and $q \leq 10$. $\square$

We write $\theta = \log \frac{26}{23} - B = 10\lambda_4 \approx 0.010264$; a spider or a tree with $\log \pi - (n-1)F^* < B$ is not a maximiser once $n \geq 91$.

3 Concave witnesses

3.1 The vertex inequality

By (1), at a vertex v with children $c_1, \dots, c_m$ and $d = m+1$, $R = \sum y_{c_i}$,

$$\ell(b_v) = \sum_i \ell(b_{c_i}) + \log \left(1 + \frac{R}{d}\right) - F^*. \quad (4)$$

Suppose we look for a bound of the form $\ell(b) \leq U(y_b)$ with U a function of the message. If the children satisfy such a bound, (4) gives $\ell(b_v) \leq \sum_i U(y_{c_i}) + \log(1 + R/d) - F^*$, and to close the induction we need the right side to be at most $U(1/(d+R))$. The sum $\sum_i U(y_{c_i})$ is the obstacle: it depends on all the children. If U is concave it is at most $mU(\bar{y})$ with $\bar{y} = R/m$, and the requirement becomes an inequality in $(m, \bar{y})$ alone.

Lemma 3.1 (concave witness, plain form). *Let $U: (0, 1] \rightarrow (-\infty, 0]$ be concave with $U(1) \geq -F^*$, and suppose that*

$$mU(\bar{y}) + \log\left(1 + \frac{m\bar{y}}{m+1}\right) - F^* \leq U\left(\frac{1}{m+1+m\bar{y}}\right) \quad \text{for all integers } m \geq 1 \text{ and all } \bar{y} \in (0, 1]. \quad (5)$$

Then $\ell(b) \leq U(y_b) \leq 0$ for every planted branch b . If moreover $U(1) = -F^$, equality holds in (5) only at $(m, \bar{y}) = (1, 1)$ and $(5, \frac{1}{3})$, and U has a corner at $\frac{1}{3}$, then $\ell(b) = U(y_b)$ only for a leaf, a cherry or A_5 , and $\ell(b) = 0$ only for $b = A_5$.*

Proof. Induction on $|b|$; a leaf has $\ell = -F^* \leq U(1)$. At a root with $m \geq 1$ children of messages y_i and mean $\bar{y}$, by (4), the induction hypothesis, Jensen and (5),

$$\ell(b) \leq \sum_i U(y_i) + \log\left(1 + \frac{m\bar{y}}{m+1}\right) - F^* \leq mU(\bar{y}) + \log\left(1 + \frac{m\bar{y}}{m+1}\right) - F^* \leq U(y_b).$$

For the equality statement, $U(\frac{1}{3}) \geq \ell(\text{cherry}) = -\varepsilon_0$ by the first part, so equality in (5) at $(5, \frac{1}{3})$ gives $U(y^*) \geq \ell(A_5) = 0$, hence $U(y^*) = 0$ and $U(\frac{1}{3}) = -\varepsilon_0$. Call b tight if $\ell(b) = U(y_b)$; leaves, cherries and A_5 are tight. If b is tight with $m \geq 1$ children, all three inequalities above are equalities: every child is tight, Jensen is tight, and $(m, \bar{y})$ is one of the two equality points. If $(m, \bar{y}) = (1, 1)$ the child is a leaf and b is a cherry. If $(m, \bar{y}) = (5, \frac{1}{3})$, every child is by induction a leaf, a cherry or an A_5 , with message 1, $\frac{1}{3}$ or y^* ; because U has a corner at $\frac{1}{3}$, a supporting line of slope strictly between the one-sided derivatives touches the graph only at $\frac{1}{3}$, so Jensen is tight only if all five messages equal $\frac{1}{3}$: the children are cherries and $b = A_5$. Finally $\ell(b) = 0$ forces $U(y_b) = 0$ and b tight, and among tight branches only A_5 has $\ell = 0$. $\square$

For the reduction to spiders we need bounds that exempt atoms, since A_5 has zero profit and cannot be charged a deficit, and that distinguish the function a branch carries from the function used to pool children. Let $\mathcal{A}$ denote the set of atoms. For a degree cap $D \in \{2, 3, \dots\}$ put $y_{\min} = 1/(2D-1)$ and $I_D = [y_{\min}, \frac{1}{2}]$; for $D = \infty$ put $I_\infty = (0, \frac{1}{2}]$. In a tree whose vertices have at most $D-1$ children, every non-leaf message lies in I_D .

Lemma 3.2 (concave witness, pooled form). *Let $D \in \{2, 3, \dots\} \cup \{\infty\}$ and $\alpha \geq 0$. Let $V: I_D \rightarrow \mathbb{R}$ be concave and $U: I_D \rightarrow \mathbb{R}$ any function with $U \leq V$. Suppose*

$$V(y(A_j)) \geq \ell(A_j) + \alpha(2j+1) \quad \text{for } 1 \leq j \leq D-1, \quad (6)$$

$$-kF^* - c\varepsilon_0 + m'V(\bar{y}) + \log\left(1 + \frac{R}{d}\right) - F^* + \alpha(1+k+2c) \leq U\left(\frac{1}{d+R}\right) \quad (7)$$

for all integers $k, c, m' \geq 0$ with $1 \leq k+c+m' \leq D-1$ other than $(1, 0, 0)$ and $(0, c, 0)$, and all $\bar{y} \in I_D$, where $d = k+c+m'+1$ and $R = k+cy_{\text{ch}} + m'\bar{y}$. Then every planted branch b that is not an atom, and all of whose vertices have at most $D-1$ children, satisfies

$$\ell(b) \leq U(y_b) - \alpha|b|.$$

Proof. Induction on $|b|$. Let the root have k leaf children, c cherry children, arm children $A_{j_1}, \dots, A_{j_a}$ and non-atom children $n_1, \dots, n_r$, and put $m' = a+r$; the root of an atom has $(k, c, m') \in \{(0, 0, 0), (1, 0, 0)\} \cup \{(0, c, 0)\}$, so (k, c, m') is not excluded. Each arm vertex has $j_i \leq D-1$ children. By (4), the exact values for leaves and cherries, (6) and the induction hypothesis with $U \leq V$,

$$\ell(b) \leq -kF^* - c\varepsilon_0 + \sum_p (V(y_p) - \alpha|p|) + \log\left(1 + \frac{R}{d}\right) - F^*,$$

the sum over the m' pooled children. Their messages lie in I_D , so Jensen gives $\sum_p V(y_p) \leq m'V(\bar{y})$, and (7) yields $\ell(b) \leq U(y_b) - \alpha(1 + k + 2c + \sum_p |p|) = U(y_b) - \alpha|b|$. $\square$

Lemma 3.2 is used three times: with $U = V$, $\alpha > 0$ and a finite cap for the rate cells (Section 5); with $\alpha = 0$, $D = \infty$ and $U \neq V$ for the envelope (Section 7); and its proof, with the exact values of the atoms, is also the root bound of Section 6.

3.2 Certificates

The inequalities (5) and (7) are, for fixed integers, inequalities in one real variable $\bar{y}$, involving a piecewise-linear function of $\bar{y}$, a logarithm of an affine function of $\bar{y}$, and the function U evaluated at $y_b = 1/(d + R)$, a Möbius function of $\bar{y}$. We use piecewise-linear U and V with rational nodes.

Lemma 3.3 (cells). *Let $[p, q]$ be an interval on which $\bar{y} \mapsto V(\bar{y})$ is affine with slope β and $\bar{y} \mapsto U(y_b(\bar{y}))$ is affine in y_b with slope β' , where $y_b = 1/(B + A\bar{y})$ with $A, B > 0$. Fix $t \in [p, q]$ and let $\Psi_t(\bar{y})$ be the left side of (7) minus its right side, with $\log(1 + R/d)$ replaced by its tangent line at t and, if $\beta' > 0$, the term $-U(y_b(\bar{y}))$ replaced by its tangent line at t as well. Then Ψ_t is convex on $[p, q]$ and dominates the difference of the two sides of (7), so $\Psi_t(p) \leq 0$ and $\Psi_t(q) \leq 0$ imply (7) on $[p, q]$, strictly except at an endpoint where Ψ_t vanishes. The same holds for (5).*

Proof. The logarithm is concave in $\bar{y}$ and lies below its tangent. The term $-U(y_b(\bar{y})) = -\alpha' - \beta'/(B + A\bar{y})$ is convex if $\beta' \leq 0$ and concave if $\beta' > 0$; in the second case it lies below its tangent. The remaining terms are affine. A convex function on an interval is at most the larger of its endpoint values, and strictly less inside unless it is constant. $\square$

A *certificate* for (7) on a cell is therefore the pair of endpoint evaluations; cells are bisected when the test fails. The endpoint values are rational combinations of $\log \frac{3}{2}$, $\log \frac{23}{18}$, of logarithms of the rational tangent arguments, and of the rational node values, since

$$F^* = \frac{1}{11} \left(5 \log \frac{3}{2} + \log \frac{23}{18} \right), \quad \varepsilon_0 = 2F^* - \log \frac{3}{2}, \quad \ell(A_j) = j \log \frac{3}{2} + \log \frac{4j+3}{3(j+1)} - (2j+1)F^*.$$

We represent every such number as a rational constant plus a formal sum $\sum_q r_q \log q$ with rational r_q and rational $q > 0$, so that identical logarithms cancel exactly: at the forced equality points the certificate value is the exact zero. For every other value we compute a rational upper bound from enclosures $\log q \in [L_q, H_q]$ obtained from $\log q = 2 \sum_{k \geq 0} t^{2k+1}/(2k+1)$, $t = (q-1)/(q+1)$, with the tail bounded by $2t^{2K+1}/((2K+1)(1-t^2))$; $K = 40$ terms give width below 10^{-25} for $q \leq 2$, and larger q are reduced by $\log q = \log(q/2) + \log 2$. The tail lemmas below use the same enclosures together with $e^x \geq 1 + x + x^2/2$ and $e^x \leq \sum_{k < K} x^k/k! + x^K/(K!(1-x/(K+1)))$ for $0 \leq x < 1$.

3.3 The tail in the plain form

Lemma 3.4 (large m). *Let U be as in Lemma 3.1, piecewise linear with nodes (y_i, U_i) , with $U \equiv 0$ on $(0, y_0]$, and let $y^\dagger = \rho - 1$. Suppose $y_0 < y^\dagger$ and that M is an integer with $1/(M+2) \leq y_0$ and $(M+1)|U_i| \geq (y_i - y^\dagger)/\rho$ at every node with $y_i > y^\dagger$. Then (5) holds strictly for every $m \geq M+1$ and every $\bar{y} \in (0, 1]$.*

Proof. For $m \geq M+1$ the right side of (5) is $U(y_b)$ with $y_b < 1/(m+1) \leq y_0$, so it is 0. The left side is less than $mU(\bar{y}) + \log(1 + \bar{y}) - F^*$, which for $\bar{y} \leq y^\dagger$ is at most $mU(\bar{y}) \leq 0$. For $\bar{y} > y^\dagger$ the concavity of the logarithm gives $\log(1 + \bar{y}) \leq F^* + (\bar{y} - y^\dagger)/\rho$, so the left side is less than $h(\bar{y}) = mU(\bar{y}) + (\bar{y} - y^\dagger)/\rho$, which is piecewise linear on $[y^\dagger, 1]$ with breakpoints at $y^\dagger$ and the nodes; $h(y^\dagger) = mU(y^\dagger) \leq 0$, and at a node $h(y_i) = (m - M - 1)U_i + [(M+1)U_i + (y_i - y^\dagger)/\rho] \leq 0$. $\square$

4 The sharp ceiling

Theorem 4.1. *Let U be the piecewise-linear function with $U = 0$ on $(0, \frac{229}{1000}]$ and nodes*

$$\left(\frac{1}{4}, -\frac{1}{2500}\right), \left(\frac{27}{100}, -\frac{141}{100000}\right), \left(\frac{29}{100}, -\frac{59}{20000}\right), \left(\frac{31}{100}, -\frac{253}{50000}\right), \left(\frac{1}{3}, \log \frac{3}{2} - 2F^*\right), (1, -F^*).$$

Then U is concave and non-positive, (5) holds, and equality occurs only at $(1, 1)$ and $(5, \frac{1}{3})$. Consequently $T_b \leq \rho^{|b|}$ for every planted branch, with equality only for A_5 , and $\max_{|T|=n} \pi(T) \leq 2\rho^{n-1}$.

Proof. The slopes of U are, in order, about $-0.019, -0.051, -0.077, -0.106, -0.113, -0.298$; the last two are $(\log \frac{3}{2} - 2F^* + \frac{253}{50000})/\frac{7}{300}$ and $\frac{3}{2}(F^* - \log \frac{3}{2})$, and the decrease is checked with the enclosures. Lemma 3.4 applies with $y_0 = \frac{229}{1000}$ and $M = 43$: $y_0 < y^\dagger \approx 0.22947$ amounts to $1.229^{11} < \frac{621}{64}$, $\frac{1}{45} \leq y_0$ is clear, and the node condition $44|U_i| \geq (y_i - y^\dagger)/\rho$ holds at the five nodes above $y^\dagger$, most tightly at $\frac{1}{4}$ where $44 \cdot \frac{1}{2500} = 0.0176$ against 0.0167 . For $1 \leq m \leq 43$ the interval $[0, 1]$ is cut at the nodes of U and at the preimages of the nodes under $\bar{y} \mapsto y_b(\bar{y})$, which gives 307 cells; Lemma 3.3 with the tangent at an endpoint when that endpoint is one of the two forced points and at the midpoint otherwise, with bisection on failure, needs 339 certificates. The two forced points give the exact zero; every other endpoint value is at most $-7.7 \cdot 10^{-5}$. The corner of U at $\frac{1}{3}$, with one-sided slopes -0.113 and -0.298 straddling $-\frac{3}{23}$, is forced by the equality at $(5, \frac{1}{3})$. Lemma 3.1 gives the rest, and (2) with $1 + S/k \leq 2$ gives the last claim. $\square$

Corollary 4.2 (growth rate). $\frac{64}{621}\rho^n \leq \max_{|T|=n} \pi(T) \leq 2\rho^{n-1}$ for every $n \geq 1$.

Proof. The upper bound is Theorem 4.1. For the lower bound take a centre carrying $\lfloor (n-1)/11 \rfloor$ arms A_5 and fill the rest with cherries and at most one leaf. $\square$

5 Rate cells

The reduction to spiders for small centre degree rests on the fact that a non-atom branch of bounded degree loses profit in proportion to its size.

Theorem 5.1 (linear deficit). *For each cap $D \in \{2, \dots, 16, 23\}$ there are a rate α_D and a concave piecewise-linear $U_D: I_D \rightarrow \mathbb{R}$, listed in Appendix B, such that (6) and (7) hold with $V = U = U_D$ and $\alpha = \alpha_D$. Consequently every planted branch b that is not an atom and whose vertices have at most $D-1$ children satisfies*

$$\ell(b) \leq U_D(y_b) - \alpha_D|b| \leq \max U_D - \alpha_D|b|.$$

The rates are

D	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	23
$1/\alpha_D$	61	114	224	376	535	660	765	878	998	1194	1335	1478	1622	1766	1912	2800
certificates	6	20	48	95	163	314	643	775	1642	1708	1931	1996	3879	4371	5088	23638

Proof. Everything is finite: (6) is a list of $D-1$ inequalities, (7) with $m' = 0$ is a single evaluation, and (7) with $m' \geq 1$ is, for each of the $\binom{D+2}{3} - D-1$ admissible triples (k, c, m') , an inequality on I_D certified by Lemma 3.3 on the cells cut by the nodes of U_D and their preimages under $\bar{y} \mapsto y_b(\bar{y})$. Every certificate is strict. Lemma 3.2 gives the conclusion. $\square$

Two remarks. The node values were found by a linear programme over a grid, maximising a margin, and then rounded to rationals with denominator 10^9 ; the linear programme is untrusted and its output is verified from scratch. And U_D cannot be non-positive: (6) forces $U_D(y^*) \geq 11\alpha_D > 0$, because an exempt A_5 child costs its parent $11\alpha_D$. That is why the conclusion carries the constant $\max U_D$, which is at most 0.05 in every case; it is harmless for the use we make of it.

For comparison, [2] uses cap 23 with rate $1/3700$ for all root degrees from 6 to 23. With the rate as a free variable the linear programme reports that cap 23 admits about $1/2507$; the rates above are near the largest the method allows for each cap.

6 Root bounds and the low-degree reduction

Lemma 6.1 (root bound). *Let T be a tree on n vertices rooted at a vertex of maximum degree k , with branches $b_1, \dots, b_k$, at least one of them not an atom, and let $D \geq k$ be a cap with rate α_D and witness U_D as in Theorem 5.1. Then*

$$\log \pi(T) - (n-1)F^* \leq R_k - \alpha_D(n-1),$$

$$R_k = \max \left\{ k_\ell(\alpha_D - F^*) + c(2\alpha_D - \varepsilon_0) + m'U_D(\bar{y}) + \log \left(1 + \frac{k_\ell + cy_{\text{ch}} + m'\bar{y}}{k} \right) : \right. \\ \left. k_\ell + c + m' = k, \ m' \geq 1, \ \bar{y} \in I_D \right\}.$$

Proof. Every vertex other than the root has at most $k-1 \leq D-1$ children, so Theorem 5.1 applies to the non-atom branches: $\ell(b) + \alpha_D|b| \leq U_D(y_b)$. For an arm, (6) gives $\ell(A_j) + \alpha_D(2j+1) \leq U_D(y_j)$; a leaf has $\ell + \alpha_D = \alpha_D - F^*$ and a cherry $2\alpha_D - \varepsilon_0$. Hence, by (3) and $\sum_i |b_i| = n-1$,

$$\log \pi - (n-1)F^* = \sum_i (\ell(b_i) + \alpha_D|b_i|) + \log \left(1 + \frac{S}{k} \right) - \alpha_D(n-1),$$

and the sum is bounded by pooling arms and non-atoms with the concave U_D by Jensen. The maximum defining R_k ranges over a superset of the configurations that occur. $\square$

The inner function $\bar{y} \mapsto m'U_D(\bar{y}) + \log(1 + (k_\ell + cy_{\text{ch}} + m'\bar{y})/k)$ is concave, so its maximum on a cell is at most the maximum over the endpoints of any tangent line, and R_k is bounded rigorously by tangent lines on subcells of width at most $1/200$. Combining with Lemma 2.5: if $n \geq 91$ and $R_k - \alpha_D(n-1) < B$, then T is not a maximiser. This holds for

$$n \geq n_0(k) := 2 + \lfloor (R_k - B)/\alpha_D \rfloor.$$

Proposition 6.2. *With the cap $D = k$ for $2 \leq k \leq 16$ and $D = 23$ for $17 \leq k \leq 23$, the root bounds and thresholds are*

k	D	$R_k \leq$	$n_0(k)$	k	D	$R_k \leq$	$n_0(k)$
2	2	0.3389	15	13	13	0.2363	185
3	3	0.3157	25	14	14	0.2234	182
4	4	0.2918	42	15	15	0.2206	193
5	5	0.2754	63	16	16	0.2249	217
6	6	0.2638	82	17	23	0.1945	232
7	7	0.2554	96	18	23	0.1988	243
8	8	0.2519	108	19	23	0.2030	255
9	9	0.2456	118	20	23	0.2072	267
10	10	0.2441	133	21	23	0.2114	279
11	11	0.2384	152	22	23	0.2157	291
12	12	0.2389	170	23	23	0.2199	303

In particular, for $n \geq 303$ every tree rooted at a vertex of maximum degree $k \leq 23$ with a non-atom branch is not a maximiser.

For small k the maximum in R_k is attained by a root carrying $k - 1$ cherries and one pooled child, for large k by k pooled children at the message y^* , that is, by the all- A_5 spider, which the pooled bound cannot tell from a tree in which one branch imitates an A_5 . Since each such child costs $11\alpha_D$, one has $n_0(23) \geq 2 + \lfloor (23 \cdot 11\alpha_D + \log \frac{26}{23} - B)/\alpha_D \rfloor \approx 255 + 0.0103/\alpha_D$, which is why the threshold cannot go much below 300 this way.

7 The high-degree envelope

For a root of degree $k \geq 24$ the bonus $\log(1 + S/k)$ is nearly linear in S , and each branch can be judged by its profit plus the value of its message at a fixed price.

Theorem 7.1 (envelope). Put $\mu_{\max} = \frac{23}{624}$ and $\delta = \frac{1}{75}$. For every planted branch b and every $\mu \in [0, \mu_{\max}]$,

$$\ell(b) + \mu y_b \leq \frac{3\mu}{23} - \delta [b \text{ is not an atom}].$$

Proof. For atoms the statement is a direct check: for a leaf, $-F^* + \mu \leq \frac{3\mu}{23}$; for a cherry, $-\varepsilon_0 + \frac{\mu}{3} \leq \frac{3\mu}{23}$ holds up to $\mu = \frac{69}{14}\varepsilon_0 \approx 0.03798$, which is where the value $\mu_{\max} \approx 0.03686$ comes from; for A_j with $j \geq 5$ both $\ell(A_j) \leq 0$ and $y_j \leq y^*$; for $j \leq 4$ the left side is affine in μ and is checked at $\mu = \mu_{\max}$, most tightly for A_4 , where it is $-1.4 \cdot 10^{-5}$.

For non-atoms both sides are affine in μ , so it suffices to prove $\ell(b) \leq T(y_b)$ with

$$T(y) = -\delta - \mu_{\max} (y - y^*)^+.$$

We apply Lemma 3.2 with $D = \infty$, $\alpha = 0$, and two piecewise-linear functions on $(0, \frac{1}{2}]$ with the 26 common nodes listed in Appendix B: $U \leq T$, with $U = T$ on $(0, \frac{1}{7}]$, is the bound carried by non-atom branches, and V with $U \leq V \leq 0$ is the pool function; V is concave and satisfies (6) with $\alpha = 0$ for $j \leq 60$, while for $j > 60$, $\ell(A_j) \leq -j\varepsilon_0 + \log \frac{4}{3} - F^* < -0.38 < \min T \leq V$. Condition (7) is certified on cells for $k \leq 2$, $c \leq 39$, $1 \leq m' \leq 80$, together with the single evaluations for $m' = 0$: 250 676 certificates, all strict, the largest endpoint value being $-1.1 \cdot 10^{-3}$. The three remaining ranges are handled by hand.

Many leaves, $k \geq 3$. Then $y_b \leq \frac{1}{2k+1} \leq \frac{1}{7}$, so $U(y_b) = T(y_b) \geq T(\frac{1}{7})$, while the left side of (7) is at most $-(k+1)F^* + \log 2 \leq -4F^* + \log 2 < -0.13 < T(\frac{1}{7})$, since $m'V \leq 0$, $-c\varepsilon_0 \leq 0$ and $R \leq d$.

Many cherries, $c \geq 40$, $k \leq 2$. Then $y_b \leq \frac{1}{c+1} < y^*$, so $U(y_b) = -\delta$. For $k = 0$ all messages are at most $\frac{1}{2}$, so $R \leq d/2$ and the left side is at most $-F^* - 40\varepsilon_0 + \log \frac{3}{2} < -0.10$; for $k \in \{1, 2\}$ it is at most $-(k+1)F^* - 40\varepsilon_0 + \log 2 < -0.028$; both are below $-\delta$.

Many pooled children, $m' > 80$, $k \leq 2$, $c \leq 39$. Then $y_b < \frac{1}{82} < y^*$, so $U(y_b) = -\delta$. With $a = k + cy_{\text{ch}}$ we have $\frac{a+m'\bar{y}}{k+c+m'+1} \leq \bar{y} + \frac{a}{m'+1}$ and $\log(1+x+\eta) \leq \log(1+x) + \eta$, so the left side is at most $[-kF^* - c\varepsilon_0 + \frac{k+c/3}{m'+1}] + m'V(\bar{y}) + \log(1+\bar{y}) - F^*$, and the bracket is ≤ 0 because $\frac{1}{m'+1} \leq \min(F^*, 3\varepsilon_0)$ once $m' \geq 43$. Let $y_1 = \frac{213}{1000}$; then $\log(1+y_1) \leq F^* - \delta$. For $\bar{y} \leq y_1$ the remaining expression is at most $m'V(\bar{y}) - \delta \leq -\delta$. For $\bar{y} > y_1$, $\log(1+\bar{y}) \leq F^* - \delta + (\bar{y} - y_1)/(1+y_1)$, and $h(\bar{y}) = m'V(\bar{y}) + (\bar{y} - y_1)/(1+y_1)$ is piecewise linear on $[y_1, \frac{1}{2}]$ with $h(y_1) \leq 0$ and, at each node $y_i > y_1$, $h(y_i) = (m' - 81)V_i + [81V_i + (y_i - y_1)/(1+y_1)] \leq 0$ by the node condition $81|V_i| \geq (y_i - y_1)/(1+y_1)$ built into V ; so $h \leq 0$ there and the left side is at most $-\delta$. $\square$

It is essential that U and V differ. Near the cherry message and near $\frac{1}{2}$ the true losses of non-atom branches are about -0.2 , while T is only about -0.02 there; with $U = V = T$ the linear programme is infeasible by a wide margin, because a vertex with five cherries and one near-cherry child would be credited far too little.

8 The reduction theorem

Proof of Theorem 1.2. Let $n \geq 303$, let T be a tree on n vertices, root it at a vertex of maximum degree k , and suppose some branch is not an atom. If $k \leq 23$, Proposition 6.2 gives $\log \pi(T) - (n-1)F^* < B$ because $n \geq 303 \geq n_0(k)$. If $k \geq 24$, put $\mu = \frac{23}{26k} \leq \mu_{\max}$. The tangent of $\log(1+x)$ at $x = \frac{3}{23}$ gives $\log(1+S/k) \leq \log \frac{26}{23} - \frac{3}{26} + \mu S$, and Theorem 7.1 gives $\sum_i (\ell(b_i) + \mu y_{b_i}) \leq k \cdot \frac{3\mu}{23} - \delta = \frac{3}{26} - \delta$, so by (2)

$$\log \pi(T) - (n-1)F^* \leq \log \frac{26}{23} - \frac{1}{75} < B,$$

because $\frac{1}{75} > \theta = 10\lambda_4$. In both cases $\pi(T)$ is less than the value of the benchmark spider of Lemma 2.5, which exists since $n \geq 91$, so T is not a maximiser. $\square$

9 The optimisation over spiders

9.1 Balance and leaves

By (3) a spider is described by its multiset of children, and moving a cherry from one arm to another keeps n , the number of children and the power of $\frac{3}{2}$ in the value.

Lemma 9.1 (balance, [2]). *Let a spider have $D \geq 3$ children, among them arms A_x and A_y with $x \geq y + 2$, leaves counted as A_0 . Replacing them by A_{x-1} and A_{y+1} strictly increases π . Hence in a maximising spider with $D \geq 3$ all arm sizes, leaves included, differ by at most one.*

Proof. With R_0 the message sum of the other children and $c = D + R_0 \geq 3$, the value is proportional to $Q_c(x, y) = \alpha_x \alpha_y (c + b_x + b_y)$, and for $x = k + t + 2$, $y = k$ with $k, t \geq 0$,

$$Q_c(k+t+1, k+1) - Q_c(k+t+2, k) = \frac{(t+1)((c-3)(8k+4t+15)+3)}{9(k+1)(k+2)(k+t+2)(k+t+3)} > 0,$$

a polynomial identity in k, t, c that we checked symbolically. $\square$

Lemma 9.2 (leaf pairs). *In a spider with $D \geq 3$ children, replacing two leaves by one cherry strictly increases π .*

Proof. With R' the message sum of the other children the value is multiplied by $\frac{3}{2} \cdot \frac{D(D+R'-\frac{2}{3})}{(D-1)(D+R'+2)}$, which exceeds 1 if and only if $(D-2)^2 + R'(D+2) > 0$. $\square$

9.2 Few children

For $j \geq 0$ real put $\lambda(x) = x\varepsilon_0 + F^* - \log \frac{4x+3}{3x+3}$, so that $\lambda(j) = \lambda_j$ for integers $j \geq 1$. The function $x \mapsto \log \frac{4x+3}{3x+3}$ is concave, as its second derivative is $-(72x+63)/((4x+3)^2(3x+3)^2)$, so λ is convex; and $\lambda'(x) = \varepsilon_0 - 1/((4x+3)(x+1)) > 0$ for $x \geq 5$.

Lemma 9.3 (few children). *A spider on $n \geq 492$ vertices with $D \leq 23$ children has $\log \pi - (n-1)F^* < B$.*

Proof. Let the centre carry ℓ leaves, c cherries and $a \geq 1$ arms with $j_1 + \dots + j_a = J$, so $\ell + 2c + a + 2J = n - 1$. By convexity, $\sum_i \lambda_{j_i} \geq a\lambda(J/a)$, and since b_j is convex and decreasing, the message sum of the arms is at most $(a-1)b_1 + b_{J-a+1}$. Hence by (3)

$$\log \pi - (n-1)F^* \leq -\ell F^* - c\varepsilon_0 - a\lambda(J/a) + \log \left(1 + \frac{\ell + c/3 + (a-1)\frac{3}{7} + b_{J-a+1}}{D} \right).$$

For $n = 492$ this is a finite list of cases, and the largest value of the right side, over all (ℓ, c, a) with $\ell + c + a \leq 23$, is -0.115 , at $\ell = c = 0$, $a = 23$; since $J/a \geq 9$ there, where λ is increasing, the right side decreases when n grows. $\square$

9.3 The tangent-gap bound

Lemma 9.4 (tangent gap). *For a spider with D children put $\mu_D = \frac{23}{26D}$ and, for an atom a , $\gamma_D(a) = \frac{3\mu_D}{23} - \ell(a) - \mu_D y_a$. Then*

$$\log \pi - (n-1)F^* \leq \log \frac{26}{23} - \sum_a \gamma_D(a).$$

Proof. The tangent of $\log(1+x)$ at $\frac{3}{23}$ gives $\log(1+S/D) \leq \log \frac{26}{23} - \frac{3}{26} + \mu_D S$, and $\sum_a \frac{3\mu_D}{23} = \frac{3}{26}$. $\square$

The gaps are non-negative once $D \geq 24$; they increase with D for atoms with $y_a > y^*$ and are at least λ_j for A_j with $j \geq 5$. At $D = 24$,

$$\gamma(\text{leaf}) > 0.174, \quad \gamma(A_1) > 0.049, \quad \gamma(A_2) > 0.016, \quad \gamma(A_3) > 0.0040, \quad \gamma(A_4) \geq 0, \quad \gamma(\text{cherry}) \geq 0,$$

while $\gamma(A_j) \geq \lambda_j$ for $j \geq 6$. A maximising spider on $n \geq 91$ vertices has $\log \pi - (n-1)F^* \geq B$, hence $\sum_a \gamma_D(a) \leq \theta = 10\lambda_4 < 0.01027$.

Corollary 9.5. *Let $n \geq 492$ and let the spider maximise π over spiders on n vertices. Then $D \geq 24$, the centre carries no leaf, no A_1 , no A_2 and no A_j with $j \geq 9$, at most two A_3 , at most six A_6 , at most two A_7 and at most one A_8 ; moreover $D \geq 32$ and at most four cherries.*

Proof. $D \geq 24$ is Lemma 9.3. The exclusions follow from $\sum \gamma \leq \theta$ and the values above, using $7\lambda_6 > \theta$, $3\lambda_7 > \theta$, $2\lambda_8 > \theta$, $\lambda_9 > \theta$ and $3\gamma_{24}(A_3) > \theta$. The children other than cherries, A_4 and A_5 then have at most $14 + 78 + 30 + 17 = 139$ vertices, so $2c + 9k_4 + 11k_5 \geq n - 1 - 139 \geq 352$ and $D \geq c + k_4 + k_5 \geq 32$; finally $5\gamma_{32}(\text{cherry}) > \theta$. $\square$

Lemma 9.6 (two families). *Let $n \geq 492$ and let the spider maximise π over spiders. Then it is $C^c A_4^{k_4} A_5^{k_5}$ or $C^c A_5^{k_5} A_6^{k_6}$ with $c \leq 4$ and $k_6 \leq 6$.*

Proof. By Lemma 9.1 the arm sizes present differ by at most one. If an A_3 is present the arms are A_3 and A_4 only, with at most two A_3 ; then $9k_4 \geq n - 1 - 8 - 14 \geq 469$, so $k_4 \geq 53$ and $D \geq 53$, and $53\gamma_{53}(A_4) > \theta$, a contradiction. If an A_7 or A_8 is present the arms have at most $78 + 30 + 17 = 125$ vertices and the cherries would number at least 183. Nothing else is possible by Corollary 9.5. $\square$

9.4 Exchanges

The three exchanges below keep n and the power of $\frac{3}{2}$; each multiplies the value by an explicit factor times a ratio of bonuses, and the condition for improvement is linear in S once denominators are cleared.

Lemma 9.7 (exchanges). *Let a spider have D children with message sum S .*

- (a) *If $D \geq 25$ and $S \geq \frac{3D}{23}$, replacing a cherry and an A_4 by an A_5 strictly increases π . The factor is $\frac{115}{114} \cdot \frac{D(D+S-\frac{473}{1311}-1)}{(D-1)(D+S)}$, which exceeds 1 if and only if $S(D+114) > D(1 + \frac{115 \cdot 473}{1311} - D)$.*
- (b) *If $D \geq 34$ and $S \geq \frac{3}{23}(D-6) + \frac{2}{3}$, replacing a cherry and an A_5 by an A_6 strictly increases π . The factor is $\frac{162}{161} \cdot \frac{D(D+S-\frac{73}{207}-1)}{(D-1)(D+S)}$, which exceeds 1 if and only if $S(D+161) > D(1 + \frac{162 \cdot 73}{207} - D)$.*
- (c) *If $D \geq 26$ and $\frac{3D}{23} \leq S \leq \frac{3D}{19}$, replacing eleven A_4 by nine A_5 strictly increases π . The factor is $\frac{T(A_5)^9}{T(A_4)^{11}} \cdot \frac{D(D+S-\frac{246}{437}-2)}{(D-2)(D+S)}$ with $T(A_5)^9/T(A_4)^{11} \approx 1.011355$.*

Proof. In each case the products of atom weights change by the stated factor, the bonus changes as stated because D and S change by the stated amounts, and the criterion is a linear inequality in S whose validity on the stated ranges is a finite check in the variable D , the right side being negative for all $S \geq 0$ once $D \geq 43, 59$ and 53 respectively. $\square$

Proposition 9.8 (candidates). *Let $n \geq 492$, $s = 6(n-1) \bmod 11$ and $m = (n-1-13s)/11$. A spider maximising π over spiders on n vertices is one of*

$$X_s = A_4^{11-s} A_5^{m+2s-9} \quad (s \geq 1), \quad X_0 = A_5^m, \quad Y_s = A_6^s A_5^m \quad (s \leq 6).$$

Proof. Take the spider of Lemma 9.6. In the first family all messages are at least y^* , so $S \geq 3D/23$, and $D \geq 32$. If $c \geq 1$ and $k_4 \geq 1$, exchange (a) applies; if $c \geq 1$ and $k_4 = 0$, then $k_5 \geq 44$, $D \geq 44$, and exchange (b) applies. So $c = 0$. If $k_4 \geq 11$ then $D \geq 45$ and $3D/23 \leq S \leq 3D/19$, and exchange (c) applies. So $k_4 \leq 10$, and $9k_4 \equiv n-1 \pmod{11}$ gives $k_4 = 5(n-1) \bmod 11 = 11-s$ for $s \geq 1$ and $k_4 = 0$ for $s = 0$. In the second family $k_6 \leq 6$, so at most six children have message below y^* and $S \geq \frac{3}{23}(D-6) + \frac{2}{3}$; also $11k_5 \geq n-1-8-78$ gives $D \geq 37$. If $c \geq 1$, exchange (b) applies. So $c = 0$, and $13k_6 \equiv n-1 \pmod{11}$ gives $k_6 \equiv s$, hence $k_6 = s \leq 6$. In both cases the number of A_5 is determined by n . $\square$

9.5 The comparisons

For $1 \leq s \leq 6$ both candidates exist and must be compared; for $s \geq 7$ only X_s survives and for $s = 0$ only X_0 . Write $G_j = T(A_j)$, so $G_4 = \frac{513}{80}$, $G_5 = \frac{621}{64}$, $G_6 = \frac{6561}{448}$, and $K_s = G_6^s G_4^{s-11} G_5^{9-2s}$. By (3),

$$\frac{\pi(Y_s)}{\pi(X_s)} = K_s \cdot \frac{\left(\frac{26}{23}m + \frac{10s}{9}\right)(m+s+2)}{(m+s)\left(\frac{26}{23}(m+2s-9) + \frac{22}{19}(11-s)\right)},$$

so Y_s wins if and only if $P_s(m) > 0$, where

$$P_s(m) = K_s \left(\frac{26}{23}m + \frac{10s}{9} \right) (m+s+2) - (m+s) \left(\frac{26}{23}(m+2s-9) + \frac{22}{19}(11-s) \right),$$

a quadratic in m with leading coefficient $\frac{26}{23}(K_s - 1)$. Since $\log K_s = (11-s)\lambda_4 - s\lambda_6$, one has $K_s > 1$ exactly for $s \leq 4$.

Proposition 9.9. *For $s \in \{1, 2\}$, $P_s(m) > 0$ for all m with $n \geq 492$: Y_s wins. For $s = 3$, $P_3(m) < 0$ for $m \leq 62$ ($n \leq 722$) and $P_3(m) > 0$ for $m \geq 63$ ($n \geq 733$). For $s = 4$, $P_4(m) < 0$ for $m \leq 206$ ($n \leq 2319$) and $P_4(m) > 0$ for $m \geq 207$ ($n \geq 2330$). For $s \in \{5, 6\}$, $P_s(m) < 0$ for all m with $n \geq 492$: X_s wins. In no case does P_s vanish at an integer.*

Proof. For $s \leq 4$ the leading coefficient is positive and the vertex of the parabola lies below the first admissible m , so P_s is increasing there, and its sign is read off at the first admissible m and, for $s = 3, 4$, at the two integers around its root. For $s \geq 5$ the leading coefficient is negative and the vertex is at a negative m , so P_s is decreasing and negative from its value at the first admissible m on. All evaluations are exact rational arithmetic. $\square$

Theorem 9.10. *For every $n \geq 492$ the unique spider maximising π over spiders on n vertices is the one described in Theorem 1.1(a).*

Proof. A maximiser exists, is one of the candidates of Proposition 9.8, and Proposition 9.9 decides between X_s and Y_s strictly; every spider that is not the winner is beaten either by an exchange or by the winner. $\square$

9.6 The range $303 \leq n \leq 491$

Proposition 9.11. *For $303 \leq n \leq 491$ the unique spider maximising π over spiders on n vertices is the one listed in Appendix C.*

Proof. By Lemma 9.1 a maximising spider with $D \geq 3$ children has all arm sizes, leaves included, within one of each other; such a spider is determined by the number of cherries and the numbers of arms of the two sizes, and there are a few thousand of them for each n . A spider with $D \leq 2$ children is either a spider with $D \geq 3$ centred elsewhere or consists of two arms. All of these were evaluated exactly, as rationals, and for each n the maximum is attained by exactly one configuration. $\square$

10 Proof of the main theorem

Proof of Theorem 1.1. Let $n \geq 303$. Trees on n vertices are finitely many, so a maximiser exists, and by Theorem 1.2 every maximiser is a spider. Hence the maximum over trees equals the maximum over spiders, and the maximisers are exactly the spiders attaining it. For $n \geq 492$ this is the spider of Theorem 9.10, for $303 \leq n \leq 491$ the spider of Proposition 9.11, unique in both cases. $\square$

11 Reproducibility

All computations are in the directory `certificates/` of the repository accompanying this paper. The file `cert.py` certifies (5) for the witness of Theorem 4.1 (`spec_1.json`) and also for seven other activities in the deformation mentioned in Section 13; `rate.py` certifies (6) and (7) for the witnesses of Theorem 5.1 (`rate_1*_root.json` and `rate_1_23_root2.json`); `root.py` computes the bounds of Proposition 6.2 from them; `env.py` certifies Theorem 7.1 (`env_1.json`); `spider_check.py` performs the checks of Section 9, including the enumeration behind Proposition 9.11. The certifying code uses only Python’s exact rationals; the linear programmes that produced the node values are separate, use floating point, and are not trusted. The complete verification takes a few minutes; the largest single item, the envelope, about ten.

12 What a formalisation would have to check

The preprint [2] formalises the matching-sum formula, the cavity recursion, the realisation of rooted trees as graphs and the assembly of ranges; those parts would be reused. What is specific to this paper is the following.

- (i) The arithmetic of formal sums $r_0 + \sum_q r_q \log q$ with rational coefficients, with exact cancellation of equal arguments, and the enclosure lemma for $\log q$ from the artanh series with the stated tail; and the elementary bounds on e^x .
- (ii) Lemmas 3.1 and 3.2 (Jensen for concave piecewise-linear functions and the two inductions), Lemma 3.3 (tangent lines and convexity on a cell) and Lemma 3.4.
- (iii) For each witness: concavity and the sign conditions on the nodes, condition (6), and the list of certificates, each an inequality between formal sums; the witness files and the cell structure are data.
- (iv) The three tail arguments of Theorem 7.1 and the atom checks, all finite inequalities.
- (v) The root bounds of Proposition 6.2: the tangent-line evaluations on subcells and the arithmetic of the thresholds.
- (vi) Lemma 2.5, the convexity of λ , the finite case list of Lemma 9.3, the gap table, the balance identity, the three exchange criteria, the sign analysis of the six quadratics, and the enumeration of Proposition 9.11, which is of the same kind as the kernel sweep of [2].

Everything here is a finite computation over $\mathbb{Q}$ or a short inequality, and nothing needs floating point, well-founded recursion or large kernel evaluations; the largest certificate family, the envelope, has 250 676 items, each a comparison of two formal sums.

13 Further remarks

The representation $\pi(T) = \mathbb{E} 2^X$ of Remark 2.2 suggests the deformation

$$\pi_\lambda(T) = \mathbb{E} (1 + \lambda)^X = \sum_M \lambda^{|M|} \prod_{uv \in M} \frac{1}{\deg u \deg v}.$$

The cavity recursion becomes $T_b = (\prod T_c)(d + \lambda R)/d$, and Lemma 3.1 applies verbatim. As $\lambda \rightarrow 0$ the problem becomes the maximisation of the Randić index $R(T)$, whose extremal trees for large n are spiders of three-cherry arms [4]; the best arm has j cherries with $j = 3, 4, 5, 6, \dots$ on successive intervals of λ , $\lambda = 1$ lying inside the A_5 interval, and $j \rightarrow \infty$ as $\lambda \uparrow 1 + \sqrt{5}$. The analogue of Theorem 4.1 has been certified at $\lambda = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{3}{2}, 2, \frac{5}{2}, 3$ by the same method, and [2] raises the corresponding question for the family with weights $(\deg u \deg v)^{-\alpha}$.

The threshold 303 is a property of the method, not of the problem. Below it the preprint’s credited cells and envelope tables remain the only proof we know, and a conceptual argument that every maximiser is a spider, valid for all n , would be of independent interest.

References

- [1] R. A. Brualdi and J. L. Goldwasser, *Permanent of the Laplacian matrix of trees and bipartite graphs*, Discrete Math. **48** (1984), 1–21.
- [2] P. W. Murphy, *The maximum Laplacian ratio of a tree: a complete answer to a question of Brualdi and Goldwasser, kernel-checked in Lean*, preprint (September 2026), <https://github.com/DrMurphyIsIn/brualdi-goldwasser>.
- [3] P. Pant, *Counterexamples to a conjecture on Laplacian ratios of trees*, arXiv:2605.14176 (2026).
- [4] L. Pavlović, M. Stojanović and X. Li, *More on the best upper bound for the Randić index R_{-1} of trees*, MATCH Commun. Math. Comput. Chem. **60** (2008), 567–584.
- [5] T. Wu, X. Dong and H.-J. Lai, *Two problems on Laplacian ratios of trees*, Discrete Appl. Math. **372** (2025), 224–236.

A Comparison with the preprint

The preprint [2] and this paper share the three-layer architecture: a cavity recursion, a reduction of maximisers to spiders, and an optimisation over spiders. They differ in how the per-vertex inequalities are certified and in what ranges of n they cover.

component	preprint	this paper
matching sum, cavity recursion	Section 2–3	reused (Section 2)
sharp ceiling (Theorem 4 there)	witness depending on child count and message; 35 cells for four-child vertices, a high-degree wrap and a dispatcher	concave witness in the message alone; 339 one-variable certificates and a two-line tail (Section 4)
reduction, high degree $k \geq 24$	three per-vertex inequalities, 22 tangent certificates	envelope with two functions U, V ; 250 676 one-variable certificates, three analytic tails (Section 7)
reduction, low degree, large n	32 rate cells and 22 root cells; cap 23 with rate $1/3700$ for $k \geq 6$; threshold 492	rate cells with a cap for each $k \leq 16$ and near-maximal rates, about 46 000 one-variable certificates; root bounds by tangent lines; threshold 303 (Sections 5 and 6)
reduction, $150 \leq n \leq 491$	credited rate cells and per-size root knapsacks (45 generated files)	not needed for $n \geq 303$; $150 \leq n \leq 302$ not covered
reduction, $7 \leq n \leq 149$	Bellman envelope tables (57 files, up to 63 GB)	not covered
spiders, $n \geq 492$	structure lemma with a global comparison, then 223 polynomial certificates over 189 shapes with up to eight cherries	tangent-gap bound, balance, three exchanges, then six polynomial inequalities (Section 9)
spiders, $n \leq 491$	kernel sweep of balanced spiders (91 files)	the same enumeration, for $303 \leq n \leq 491$ only
trusted computation	Lean 4 kernel, three standard axioms	exact rational arithmetic in Python, not formalised

The certificate counts are not comparable one to one: a preprint cell is a multi-dimensional statement, a certificate here is two evaluations. The gain is in structure. The preprint’s cells are indexed by the types of all children of a vertex and need a dispatcher that sorts them; here every check is an inequality on an interval of the real line, and the same three lemmas serve the ceiling, the rate cells, the root bounds and the envelope. The price is the threshold: our root bounds cannot distinguish the all- A_5 spider from a tree with one branch imitating an A_5 , which pins the threshold near 300, whereas the preprint pushes below 150 with credited cells and to 7 with envelope tables, both of which are genuinely different, size-indexed computations.

Two smaller points. The preprint’s explanation of its threshold 492 does not reproduce from the constants displayed there ($R_k \leq 0.319$, rate $1/3700$, benchmark 0.1122 give about 765); with our own root bound at the preprint’s cap 23 and rate $1/3700$ the threshold came out as exactly 492, at root degree 6, which suggests the preprint’s bound is a per-degree computation of the same kind. And the preprint’s optimisation over spiders allows up to eight cherries in its final candidate set; the exchange of a cherry and an A_5 for an A_6 , valid from centre degree 34, removes them.

B Witness data

The ceiling. The witness of Theorem 4.1 is given in its statement. The tail lemma is applied with $M = 43$.

Rate cells. For each cap D the witness U_D is the piecewise-linear function with the nodes $(y, U_D(y))$ listed below, on $I_D = [1/(2D - 1), \frac{1}{2}]$; node values are rationals with denominator 10^9 ,

exactly as certified.

D	α_D	nodes
2	1/61	$(\frac{1}{3}, -1), (\frac{119}{354}, 8582963/500000000), (\frac{73}{177}, -1298503/250000000), (\frac{3}{7}, -9962429/1000000000),$ $(\frac{88}{177}, -8698707/250000000), (\frac{1}{2}, -1)$
3	1/114	$(\frac{1}{5}, -20404283/250000000), (\frac{121}{590}, 1970157/500000000), (\frac{3}{11}, 5882413/250000000),$ $(\frac{146}{295}, -56818331/1000000000), (\frac{1}{2}, -1)$
4	1/224	$(\frac{1}{7}, -61967949/2500000000), (\frac{123}{826}, -144067367/1000000000), (\frac{1}{5}, 6421401/2500000000),$ $(\frac{204}{413}, -13236333/2000000000), (\frac{1}{2}, -1)$
5	1/376	$(\frac{1}{9}, -430846371/10000000000), (\frac{125}{1062}, 5265061/2500000000), (\frac{3}{19}, 11954873/5000000000),$ $(\frac{517}{1062}, -30452559/5000000000), (\frac{1}{2}, -1)$
6	1/535	$(\frac{1}{11}, 12411793/5000000000), (\frac{3}{23}, 5390187/2500000000), (\frac{311}{649}, -38955397/10000000000),$ $(\frac{320}{649}, -4138559/1000000000), (\frac{1}{2}, -1)$
7	1/660	$(\frac{1}{13}, 21862101/10000000000), (\frac{3}{23}, 17666667/10000000000), (\frac{24}{59}, -4628473/2500000000),$ $(\frac{635}{1534}, -9835903/5000000000), (\frac{378}{767}, -39130363/10000000000), (\frac{1}{2}, -1)$
8	1/765	$(\frac{1}{15}, 22486803/10000000000), (\frac{131}{1770}, 10861737/5000000000), (\frac{98}{885}, 8912137/5000000000),$ $(\frac{3}{23}, 3075817/2000000000), (\frac{703}{1770}, -2498367/1250000000), (\frac{3}{7}, -12191023/5000000000),$ $(\frac{436}{885}, -40305967/10000000000), (\frac{1}{2}, -1)$
9	1/878	$(\frac{1}{17}, 8362491/5000000000), (\frac{3}{23}, 6764237/5000000000), (\frac{3}{19}, 2044829/2000000000),$ $(\frac{883}{2006}, -15682463/5000000000), (\frac{943}{2006}, -18022991/5000000000), (\frac{494}{1003}, -41491281/10000000000),$ $(\frac{1}{2}, -1)$
10	1/998	$(\frac{1}{19}, -1), (\frac{169}{2242}, -45456707/10000000000), (\frac{1}{13}, 3369549/2000000000),$ $(\frac{237}{2242}, 14320421/10000000000), (\frac{144}{1121}, 6157153/5000000000), (\frac{161}{1121}, 10815297/10000000000),$ $(\frac{849}{2242}, -19218531/10000000000), (\frac{433}{1121}, -21182099/10000000000), (\frac{552}{1121}, -10445599/2000000000),$ $(\frac{1}{2}, -1)$
11	1/1194	$(\frac{1}{21}, -1), (\frac{3}{43}, 4987031/5000000000), (\frac{25}{354}, 15397069/10000000000), (\frac{3}{19}, 53213/7812500),$ $(\frac{139}{354}, -23242437/10000000000), (\frac{337}{826}, -322069/125000000), (\frac{610}{1239}, -48057741/10000000000),$ $(\frac{1}{2}, -1)$
12	1/1335	$(\frac{1}{23}, -1), (\frac{3}{47}, -3604431/5000000000), (\frac{181}{2714}, 14525161/10000000000), (\frac{143}{1357}, 2882379/2500000000),$ $(\frac{3}{19}, 1503787/2500000000), (\frac{668}{1357}, -9972027/2000000000), (\frac{1}{2}, -1)$
13	1/1478	$(\frac{1}{25}, 15150959/10000000000), (\frac{187}{2950}, 13447149/10000000000), (\frac{3}{23}, 774249/10000000000),$ $(\frac{3}{19}, 1072577/2000000000), (\frac{726}{1475}, -49351689/10000000000), (\frac{1}{2}, -1)$
14	1/1622	$(\frac{1}{27}, -43271891/10000000000), (\frac{109}{1593}, 10301417/10000000000), (\frac{3}{43}, 11135043/10000000000),$ $(\frac{3}{23}, 7081751/10000000000), (\frac{3}{19}, 4822281/10000000000), (\frac{1393}{3186}, -15878223/5000000000),$ $(\frac{734}{1593}, -4381163/1250000000), (\frac{784}{1593}, -42762461/10000000000), (\frac{1}{2}, -1)$
15	1/1766	$(\frac{1}{29}, -1), (\frac{3}{59}, -13366077/5000000000), (\frac{3}{55}, -8963971/10000000000), (\frac{199}{3422}, 254847/5000000000),$ $(\frac{3}{23}, 3264383/5000000000), (\frac{3}{19}, 2184919/5000000000), (\frac{842}{1711}, -48577783/10000000000), (\frac{1}{2}, -1)$
16	1/1912	$(\frac{1}{31}, -20944069/2500000000), (\frac{205}{3658}, 509121/1250000000), (\frac{3}{43}, 1229839/1250000000),$ $(\frac{3}{23}, 3026569/5000000000), (\frac{3}{19}, 248793/625000000), (\frac{1771}{3658}, -5879367/1250000000),$ $(\frac{900}{1829}, -4277927/156250000), (\frac{1}{2}, -1)$
23	1/2800	$(\frac{1}{45}, -1), (\frac{3}{91}, -83021367/10000000000), (\frac{1}{29}, 418617/1250000000), (\frac{34}{885}, 57119/1250000000),$ $(\frac{3}{23}, 4228571/10000000000), (\frac{3}{19}, 2487861/10000000000), (\frac{1}{5}, -941099/2500000000),$ $(\frac{3}{11}, -9622141/5000000000), (\frac{3}{7}, -58771329/10000000000), (\frac{2483}{5310}, -67424543/5000000000),$ $(\frac{1306}{2655}, -92558333/5000000000), (\frac{1}{2}, -1)$

The envelope. The two functions of Theorem 7.1 have the common nodes below; $U = T$ at nodes up to $\frac{1}{7}$, V is the pool function, and the tail is applied with $M = 80$ and $y_1 = \frac{213}{1000}$.

y	$U(y)$	$V(y)$	y	$U(y)$	$V(y)$
1/50	-1/75	0	6/25	-3474359/200000000	-2989667/250000000
1/20	-1/75	0	1/4	-3548077/200000000	-3326809/250000000
2/25	-1/75	0	13/50	-724359/40000000	-3663951/250000000
1/10	-1/75	0	7/25	-9423077/500000000	-867647/500000000
3/25	-1/75	0	3/10	-20050077/1000000000	-5012519/250000000
3/23	-1/75	0	8/25	-22747213/1000000000	-5686803/250000000
1/7	-251/18200	0	1/3	-24545303/1000000000	-18408977/750000000
4/25	-14423077/1000000000	-292531/250000000	7/20	-1	-6698229/250000000
9/50	-15160257/1000000000	-193363/500000000	19/50	-30838621/1000000000	-1541931/500000000
1/5	-3974359/250000000	-1641099/250000000	41/100	-1395373/40000000	-8721081/250000000
213/1000	-16376603/1000000000	-5198459/625000000	11/25	-39176633/1000000000	-4897079/125000000
11/50	-2079327/125000000	-2315383/250000000	47/100	-22491939/500000000	-44983877/1000000000
23/100	-8501603/500000000	-106101/100000000	1/2	-25632789/500000000	-25632789/500000000

C The maximisers for $303 \leq n \leq 491$

Notation: C^c is c cherries at the centre and A_j^m is m arms with j cherries. Each entry is the unique maximiser over spiders, by exact enumeration, and by Theorem 1.2 over all trees.

n	maximiser	n	maximiser	n	maximiser	n	maximiser
303	$A_5^{25} A_4^3$	351	$A_5^{31} A_4$	399	$A_6 A_5^{35}$	447	$A_5^{34} A_4^8$
304	$A_5^{21} A_4^8$	352	$A_5^{27} A_4^6$	400	$A_5^{33} A_4^4$	448	$A_5^{39} A_4^2$
305	$A_5^{26} A_4^2$	353	A_5^{32}	401	$A_5^{29} A_4^9$	449	$A_5^{35} A_4^7$
306	$A_5^{22} A_4^7$	354	$A_5^{28} A_4^5$	402	$A_5^{34} A_4^3$	450	$A_5^{40} A_4$
307	$A_5^{27} A_4$	355	$A_6 A_5^{31}$	403	$A_5^{30} A_4^8$	451	$A_5^{36} A_4^6$
308	$A_5^{23} A_4^6$	356	$A_5^{29} A_4^4$	404	$A_5^{35} A_4^2$	452	A_5^{41}
309	A_5^{28}	357	$A_5^{25} A_4^9$	405	$A_5^{31} A_4^7$	453	$A_5^{37} A_4^5$
310	$A_5^{24} A_4^5$	358	$A_5^{30} A_4^3$	406	$A_5^{36} A_4$	454	$A_6 A_5^{40}$
311	$C^1 A_5^{28}$	359	$A_5^{26} A_4^8$	407	$A_5^{32} A_4^6$	455	$A_5^{38} A_4^4$
312	$A_5^{25} A_4^4$	360	$A_5^{31} A_4^2$	408	A_5^{37}	456	$A_6^2 A_5^{39}$
313	$A_5^{21} A_4^9$	361	$A_5^{27} A_4^7$	409	$A_5^{33} A_4^5$	457	$A_5^{39} A_4^3$
314	$A_5^{26} A_4^3$	362	$A_5^{32} A_4$	410	$A_6 A_5^{36}$	458	$A_5^{35} A_4^8$
315	$A_5^{22} A_4^8$	363	$A_5^{28} A_4^6$	411	$A_5^{34} A_4^4$	459	$A_5^{40} A_4^2$
316	$A_5^{27} A_4^2$	364	A_5^{33}	412	$A_5^{30} A_4^9$	460	$A_5^{36} A_4^7$
317	$A_5^{23} A_4^7$	365	$A_5^{29} A_4^5$	413	$A_5^{35} A_4^3$	461	$A_5^{41} A_4$
318	$A_5^{28} A_4$	366	$A_6 A_5^{32}$	414	$A_5^{31} A_4^8$	462	$A_5^{37} A_4^6$
319	$A_5^{24} A_4^6$	367	$A_5^{30} A_4^4$	415	$A_5^{36} A_4^2$	463	A_5^{42}
320	A_5^{29}	368	$A_5^{26} A_4^9$	416	$A_5^{32} A_4^7$	464	$A_5^{38} A_4^5$
321	$A_5^{25} A_4^5$	369	$A_5^{31} A_4^3$	417	$A_5^{37} A_4$	465	$A_6 A_5^{41}$
322	$C^1 A_5^{29}$	370	$A_5^{27} A_4^8$	418	$A_5^{33} A_4^6$	466	$A_5^{39} A_4^4$
323	$A_5^{26} A_4^4$	371	$A_5^{32} A_4^2$	419	A_5^{38}	467	$A_6^2 A_5^{40}$
324	$A_5^{22} A_4^9$	372	$A_5^{28} A_4^7$	420	$A_5^{34} A_4^5$	468	$A_5^{40} A_4^3$
325	$A_5^{27} A_4^3$	373	$A_5^{33} A_4$	421	$A_6 A_5^{37}$	469	$A_5^{36} A_4^8$
326	$A_5^{23} A_4^8$	374	$A_5^{29} A_4^6$	422	$A_5^{35} A_4^4$	470	$A_5^{41} A_4^2$
327	$A_5^{28} A_4^2$	375	A_5^{34}	423	$A_5^{31} A_4^9$	471	$A_5^{37} A_4^7$
328	$A_5^{24} A_4^7$	376	$A_5^{30} A_4^5$	424	$A_5^{36} A_4^3$	472	$A_5^{42} A_4$
329	$A_5^{29} A_4$	377	$A_6 A_5^{33}$	425	$A_5^{32} A_4^8$	473	$A_5^{38} A_4^6$
330	$A_5^{25} A_4^6$	378	$A_5^{31} A_4^4$	426	$A_5^{37} A_4^2$	474	A_5^{43}
331	A_5^{30}	379	$A_5^{27} A_4^9$	427	$A_5^{33} A_4^7$	475	$A_5^{39} A_4^5$
332	$A_5^{26} A_4^5$	380	$A_5^{32} A_4^3$	428	$A_5^{38} A_4$	476	$A_6 A_5^{42}$
333	$C^1 A_5^{30}$	381	$A_5^{28} A_4^8$	429	$A_5^{34} A_4^6$	477	$A_5^{40} A_4^4$
334	$A_5^{27} A_4^4$	382	$A_5^{33} A_4^2$	430	A_5^{39}	478	$A_6^2 A_5^{41}$
335	$A_5^{23} A_4^9$	383	$A_5^{29} A_4^7$	431	$A_5^{35} A_4^5$	479	$A_5^{41} A_4^3$
336	$A_5^{28} A_4^3$	384	$A_5^{34} A_4$	432	$A_6 A_5^{38}$	480	$A_5^{37} A_4^8$
337	$A_5^{24} A_4^8$	385	$A_5^{30} A_4^6$	433	$A_5^{36} A_4^4$	481	$A_5^{42} A_4^2$
338	$A_5^{29} A_4^2$	386	A_5^{35}	434	$A_6^2 A_5^{37}$	482	$A_5^{38} A_4^7$
339	$A_5^{25} A_4^7$	387	$A_5^{31} A_4^5$	435	$A_5^{37} A_4^3$	483	$A_5^{43} A_4$
340	$A_5^{30} A_4$	388	$A_6 A_5^{34}$	436	$A_5^{33} A_4^8$	484	$A_5^{39} A_4^6$
341	$A_5^{26} A_4^6$	389	$A_5^{32} A_4^4$	437	$A_5^{38} A_4^2$	485	A_5^{44}
342	A_5^{31}	390	$A_5^{28} A_4^9$	438	$A_5^{34} A_4^7$	486	$A_5^{40} A_4^5$
343	$A_5^{27} A_4^5$	391	$A_5^{33} A_4^3$	439	$A_5^{39} A_4$	487	$A_6 A_5^{43}$
344	$A_6 A_5^{30}$	392	$A_5^{29} A_4^8$	440	$A_5^{35} A_4^6$	488	$A_5^{41} A_4^4$
345	$A_5^{28} A_4^4$	393	$A_5^{34} A_4^2$	441	A_5^{40}	489	$A_6^2 A_5^{42}$
346	$A_5^{24} A_4^9$	394	$A_5^{30} A_4^7$	442	$A_5^{36} A_4^5$	490	$A_5^{42} A_4^3$
347	$A_5^{29} A_4^3$	395	$A_5^{35} A_4$	443	$A_6 A_5^{39}$	491	$A_5^{38} A_4^8$
348	$A_5^{25} A_4^8$	396	$A_5^{31} A_4^6$	444	$A_5^{37} A_4^4$		
349	$A_5^{30} A_4^2$	397	A_5^{36}	445	$A_6^2 A_5^{38}$		
350	$A_5^{26} A_4^7$	398	$A_5^{32} A_4^5$	446	$A_5^{38} A_4^3$		